```
2dlist = [list1, list2, list3]
```

A list made up of lists

	Sex	Race	Height	Income	Marital Status	Years of Educ.
R1001	M	1	70	50	1	12
R1002	M	2	72	100	2	20
R1003	F	1	55	250	1	16
R1004	M	2	65	20	2	16
R1005	F	1	60	10	3	12
R1006	M	1	68	30	1	9
R1007	F	5	66	25	2	21
R1008	F	4	61	43	1	18
R1009	M	1	69	67	1	12

```python
fruits = ["apple", "orange", "banana", "coconut"]
vegetables = ["celery", "carrots", "potatoes"]
meats = ["chicken", "fish", "turkey"]

groceries = [fruits, vegetables, meats]

fruits[0]
print(fruits)
```

main.py

```python

fruits = ["apple", "orange", "banana", "coconut"]
vegetables = ["celery", "carrots", "potatoes"]
meats = ["chicken", "fish", "turkey"]

groceries = [fruits, vegetables, meats]

fruits[0] = "pineapple"
print(fruits)
```

```python

fruits =      ["apple", "orange", "banana", "coconut"]
vegetables = ["celery", "carrots", "potatoes"]
meats =      ["chicken", "fish", "turkey"]

groceries = [fruits, vegetables, meats]

print(groceries[0])
```

This returns an entire row

```
fruits =      ["apple", "orange", "banana", "coconut"]
vegetables = ["celery", "carrots", "potatoes"]
meats =       ["chicken", "fish", "turkey"]

groceries = [fruits, vegetables, meats]

print(groceries[0][1])
```

main.py

```python

fruits = ["apple", "orange", "banana", "coconut"]
vegetables = ["celery", "carrots", "potatoes"]
meats = ["chicken", "fish", "turkey"]

groceries = [fruits, vegetables, meats]

print(groceries[1][1])
```

```python
fruits = ["apple", "orange", "banana", "coconut"]
vegetables = ["celery", "carrots", "potatoes"]
meats = ["chicken", "fish", "turkey"]

groceries = [fruits, vegetables, meats]

print(groceries[2][3])
```

```python

fruits = ["apple", "orange", "banana", "coconut"]
vegetables = ["celery", "carrots", "potatoes"]
meats = ["chicken", "fish", "turkey"]

groceries = [fruits, vegetables, meats]

print(groceries[2][1])
```

```python

fruits =      ["apple", "orange", "banana", "coconut"]
vegetables = ["celery", "carrots", "potatoes"]
meats =      ["chicken", "fish", "turkey"]

groceries = [fruits, vegetables, meats]

print(groceries[0][0])
```

```python
groceries = [["apple", "orange", "banana", "coconut"],
             ["celery", "carrots", "potatoes"],
             ["chicken", "fish", "turkey"]]

print(groceries[0][0])
```

```python
groceries = [["apple", "orange", "banana", "coconut"],
             ["celery", "carrots", "potatoes"],
             ["chicken", "fish", "turkey"]]

for collection in groceries:
    print(collection)
```

This iterates over the rows

```
['apple', 'orange', 'banana', 'coconut']
['celery', 'carrots', 'potatoes']
['chicken', 'fish', 'turkey']
```

main.py

```python
groceries = [["apple", "orange", "banana", "coconut"],
             ["celery", "carrots", "potatoes"],
             ["chicken", "fish", "turkey"]]

for collection in groceries:
    for food in collection:
        print(food)
```

This iterates over the elements in each row

for collection in groceries › for food in collection

```
Users\HP\PycharmProjects\practice\venv\bin\python.exe C:/Users/HP/PycharmProjects/practice/main.py
ple
ange
nana
conut
```

practice main.py

```python
groceries = [["apple", "orange", "banana", "coconut"],
             ["celery", "carrots", "potatoes"],
             ["chicken", "fish", "turkey"]]

for collection in groceries:
    for food in collection:
        print(food, end=" ")
    print()
```

for collection in groceries

Run: main

```
apple orange banana coconut celery carrots potatoes chicken fish turkey
```

practice C:\Users\HP\Pych
venv
main.py
External Libraries
Scratches and Consoles

```python
num_pad = ((1, 2, 3),
           (4, 5, 6),
           (7, 8, 9),
           ("*", 0, "#"))

for row in num_pad:
    print(row)
```

for row in num_pad

main

```
C:\Users\HP\PycharmProjects\practice\venv\bin\python.exe C:/Users/HP/Py
(1, 2, 3)
(4, 5, 6)
(7, 8, 9)
('*', 0, '#')
```

practice C:\Users\HP\PyCr

venv

main.py

External Libraries

Scratches and Consoles

```python
num_pad = ((1, 2, 3),
           (4, 5, 6),
           (7, 8, 9),
           ("*", 0, "#"))

for row in num_pad:
    for num in row:
        print(num, end=" ")
```

for row in num_pad › for num in row

un: main

```
*
0
#
```

venv
main.py
External Libraries
Scratches and Consoles

```python
num_pad = ((1, 2, 3),
           (4, 5, 6),
           (7, 8, 9),
           ("*", 0, "#"))

for row in num_pad:
    for num in row:
        print(num, end=" ")
    print()
```

for row in num_pad

Run: main

```
C:\Users\HP\PycharmProjects\practice\venv\bin\python.exe C:/Users
1 2 3
4 5 6
7 8 9
* 0 #
```